A. Hypothesis Testing (10 Problems)

1. A **battery company claims** that its batteries last **50 hours on average**. A consumer group tests **30 batteries** and finds an average of **48 hours** with a standard deviation of **5 hours**. At $\alpha = 0.05$, test if the company's claim is true.
2. A university claims that the **average GPA** of its students is **3.5**. A researcher collects **40 student GPAs** and finds an average of **3.3** with a standard deviation of **0.6**. Test if the university's claim is correct at $\alpha = 0.01$.
3. A pharmaceutical company introduces a **new drug** that is supposed to lower **blood pressure by 10 mmHg**. A study on **25 patients** shows an average reduction of **8.5 mmHg** with a standard deviation of **3.2 mmHg**. Test at $\alpha = 0.05$ whether the drug is effective.
4. A call center manager claims that the **average customer wait time** is **less than 5 minutes**. A random sample of **20 calls** shows an average wait time of **4.5 minutes** with a standard deviation of **1.2 minutes**. Test at $\alpha = 0.05$.
5. A **sports drink company** claims that its new energy drink improves **running speed**. A test on **15 athletes** shows an average speed increase of **0.8 seconds**, with a standard deviation of **0.3 seconds**. Test if the drink has a significant effect at $\alpha = 0.01$.
6. A study suggests that people should get at least **7 hours of sleep** per night. A survey of **35 people** finds an average sleep time of **6.8 hours**, with a standard deviation of **1.5 hours**. Test if the population sleeps less than 7 hours at $\alpha = 0.05$.
7. A company claims that the **average monthly sales** of a product are **$10,000**. A sample of **50 months** shows an average of **$9,500** with a standard deviation of **$1,200**. Test if the company's claim is valid at $\alpha = 0.05$.
8. A researcher claims that the **average weight of newborns** is **3.5 kg**. A hospital records the weights of **30 newborns** and finds an average of **3.4 kg** with a standard deviation of **0.4 kg**. Test at $\alpha = 0.05$.
9. A **college professor** believes that students who study **at night** perform better than those who study during the **day**. A test on **30 students** shows an average grade of **85%** for night-study students and **80%** for day-study students, with standard deviations of **5%** and **6%**, respectively. Test if night studying significantly improves performance at $\alpha = 0.05$.
10. A **restaurant** claims that their **average customer rating** is **4.5 stars**. A survey of **40 customers** results in an average rating of **4.3 stars**, with a standard deviation of **0.5 stars**. Test the claim at $\alpha = 0.05$.

B. Z-Test (10 Problems)

1. A **machine fills juice bottles** and is supposed to dispense **500ml** per bottle. A sample of **50 bottles** shows an average of **495ml** with a standard deviation of **10ml**. Test at $\alpha = 0.05$.
2. A **company claims** that its employees work **40 hours per week** on average. A survey of **100 employees** finds an average of **38 hours**, with a known standard deviation of **5 hours**. Test at

$\alpha = 0.05$.

3. A **bank** claims that the **average transaction time** at an ATM is **3 minutes**. A sample of **60 customers** shows an average transaction time of **3.2 minutes**, with a standard deviation of **0.5 minutes**. Test at $\alpha = 0.01$.
4. A **university claims** that the **average student spends $500 per semester** on textbooks. A sample of **80 students** shows an average of **$480**, with a standard deviation of **$50**. Test at $\alpha = 0.05$.
5. A **new smartphone battery** is advertised to last **24 hours**. A test on **75 phones** shows an average battery life of **23.5 hours**, with a standard deviation of **1.8 hours**. Test at $\alpha = 0.05$.
6. A **supermarket** claims that the **average checkout time** is **5 minutes**. A survey of **90 customers** finds an average checkout time of **5.4 minutes**, with a standard deviation of **1.2 minutes**. Test at $\alpha = 0.05$.
7. A **fast-food chain** claims that the **average burger weight** is **150 grams**. A sample of **50 burgers** shows an average weight of **148 grams**, with a standard deviation of **5 grams**. Test at $\alpha = 0.05$.
8. A **car rental company** claims that the **average rental duration** is **7 days**. A sample of **40 customers** finds an average rental of **6.5 days**, with a standard deviation of **2 days**. Test at $\alpha = 0.05$.
9. A **hospital** claims that the **average patient wait time** in an emergency room is **20 minutes**. A sample of **120 patients** finds an average wait time of **22 minutes**, with a standard deviation of **4 minutes**. Test at $\alpha = 0.01$.
10. A **company claims** that its product has a **defect rate of 5%**. A test on **200 products** finds **15 defective items**. Test at $\alpha = 0.05$.

C. T-Test (10 Problems)

- Repeat the **above problems for small sample sizes** ($n < 30$), where population **standard deviation is unknown**.

D. Chi-Square Test (10 Problems)

1. A **survey** records the **favorite pizza type** of 200 customers (**Goodness of Fit Test**).
2. A **study examines** if **gender is related to shopping preference** (**Independence Test**).
3. A **school wants to know** if students' **preferred study times** are evenly distributed.
4. A **survey checks** if different age groups **prefer different smartphone brands**.
5. A **hospital examines** whether patient **recovery rates depend on medication type**.
6. A **company studies** if customer satisfaction **varies by service type**.

7. A **city government** tests if traffic accidents **are equally distributed across days**.
8. A **restaurant** investigates if meal preferences **differ by gender**.
9. A **university analyzes** if student majors **affect job placement rates**.
10. A **movie studio** examines whether people **prefer different genres equally**